

Supplementary Materials for **SpaTemporal**

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1 Supplementary Figures

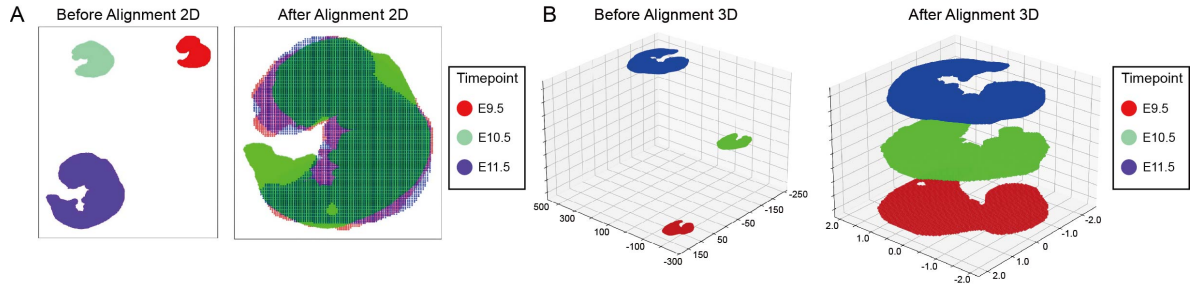


Fig. S1 Comparison between (A) 2D and (B) 3D visualization of mouse embryo slices (E9.5, E10.5, E11.5) before and after SpaTemporal alignment.

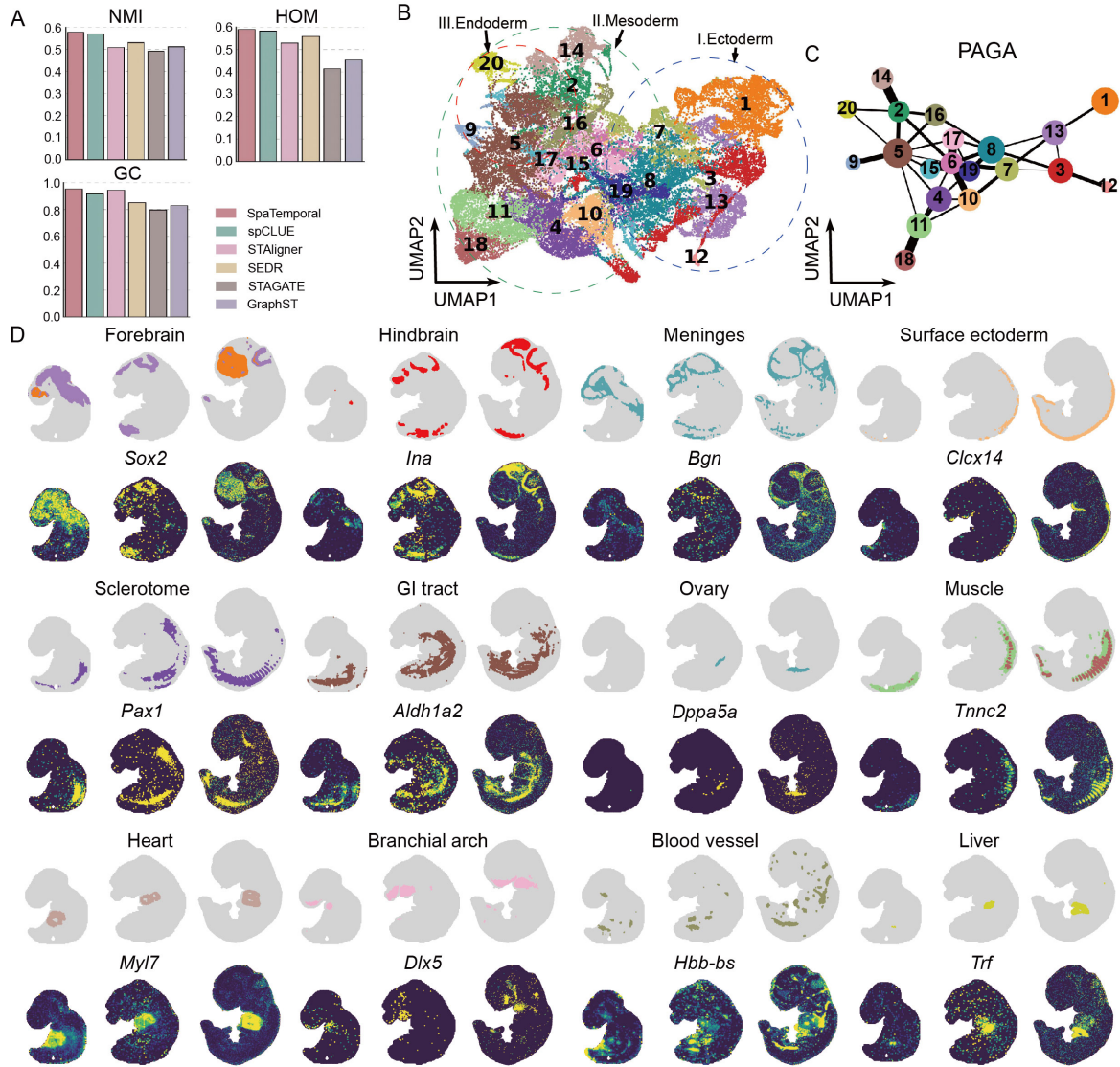


Fig. S2 SpaTemporal accurately identifies continuous structures across mouse embryonic developmental stages and delineates tissues derived from different germ layers. **(A)** Bar plots represent the mean NMI, HOM and GC scores of SpaTemporal and baseline methods across multiple developmental time points. **(B)** SpaTemporal delineates tissues originating from different germ layers. **(C)** Developmental trajectories and connectivity inferred by PAGA. **(D)** SpaTemporal accurately identifies the spatial domains that align well with anatomical structures and marker gene patterns.

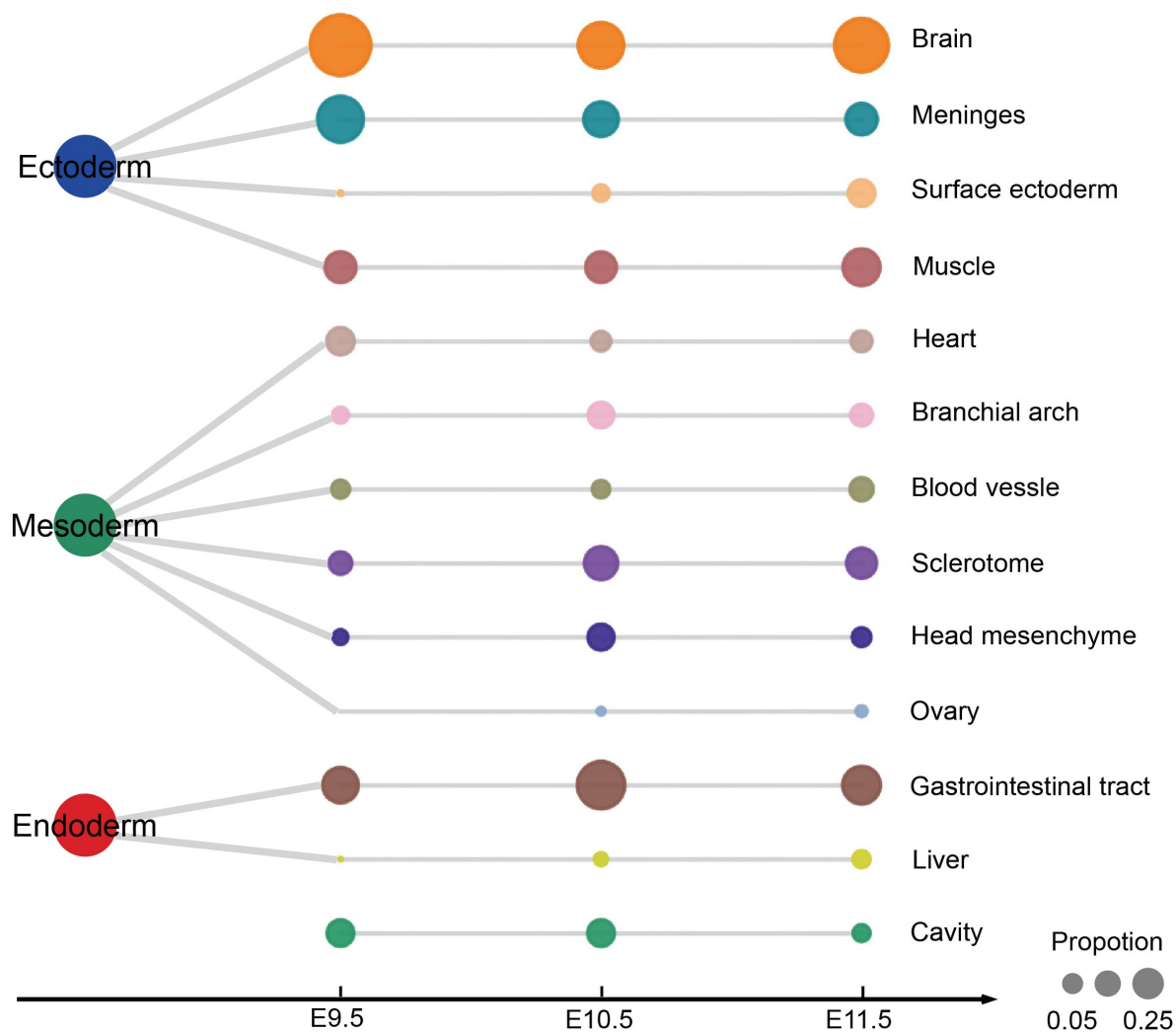


Fig. S3 Developmental dynamics of tissue substructures shown by a directed acyclic graph.

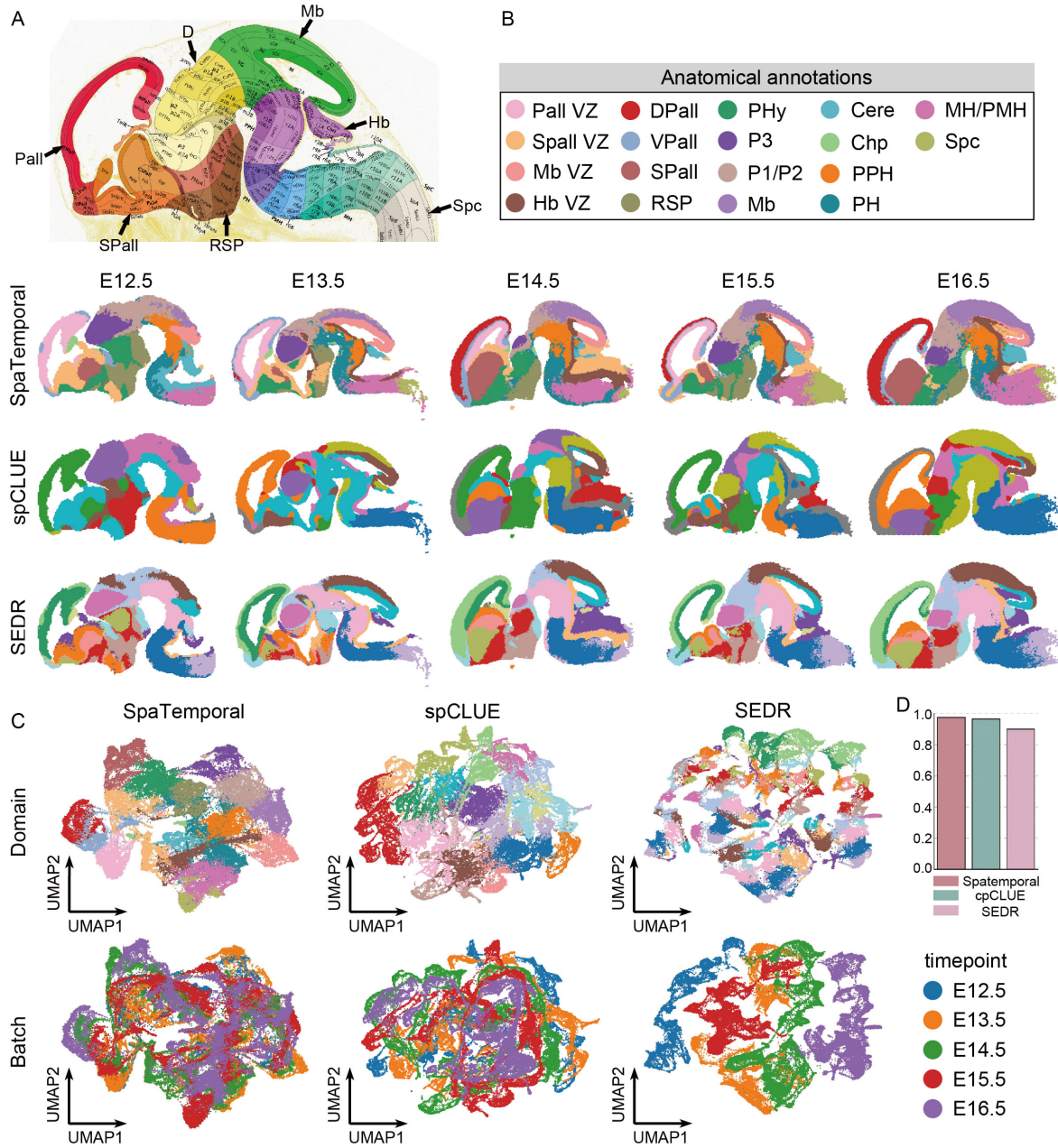


Fig. S4 Anatomical regional annotations and the spatial domain identification results of SpaTemporal and Baselines in the developing mouse brain. **(A)** Comprehensive anatomical region annotations were obtained from the Allen Brain Atlas at the E13.5 developmental stage to serve as the ground truth reference. **(B)** Spatial domain identification by SpaTemporal and two baseline methods in the developing mouse brain, complemented by anatomical annotations from the Allen Brain Atlas. **(C)** UMAP visualization of embeddings colored by identified domains (top) and batch corrected (bottom). **(D)** Bar plot of GC scores evaluating the data integration of SpaTemporal, spCLUE, and SEDR.

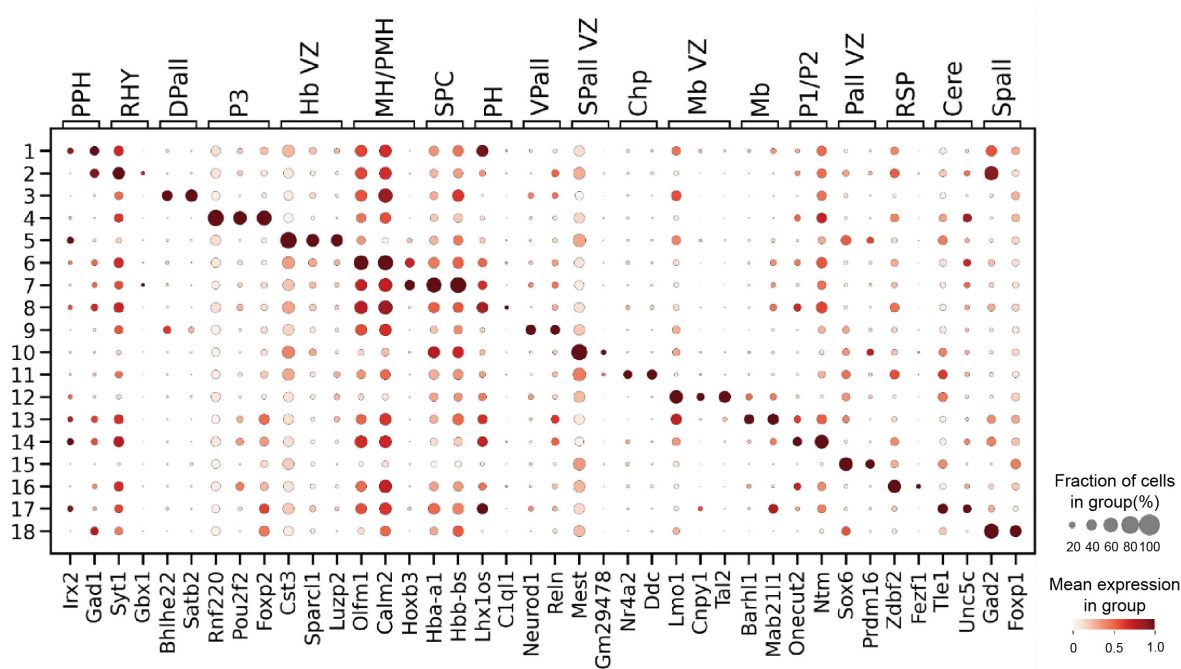


Fig. S5 Dot plots of marker genes for all spatial domains during mouse brain development.

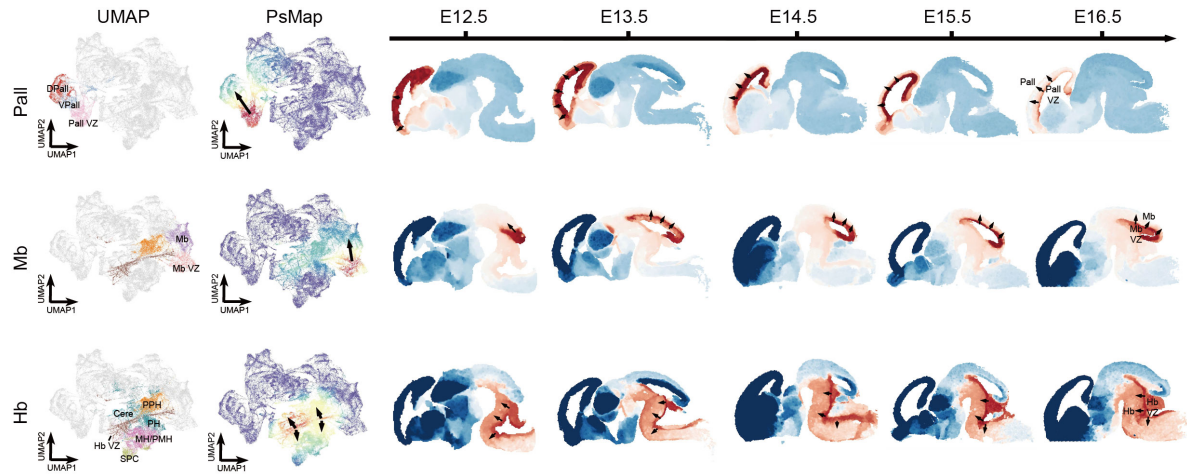


Fig. S6 Spatial visualization of PsMap derived from VZ for specific spatial domains.

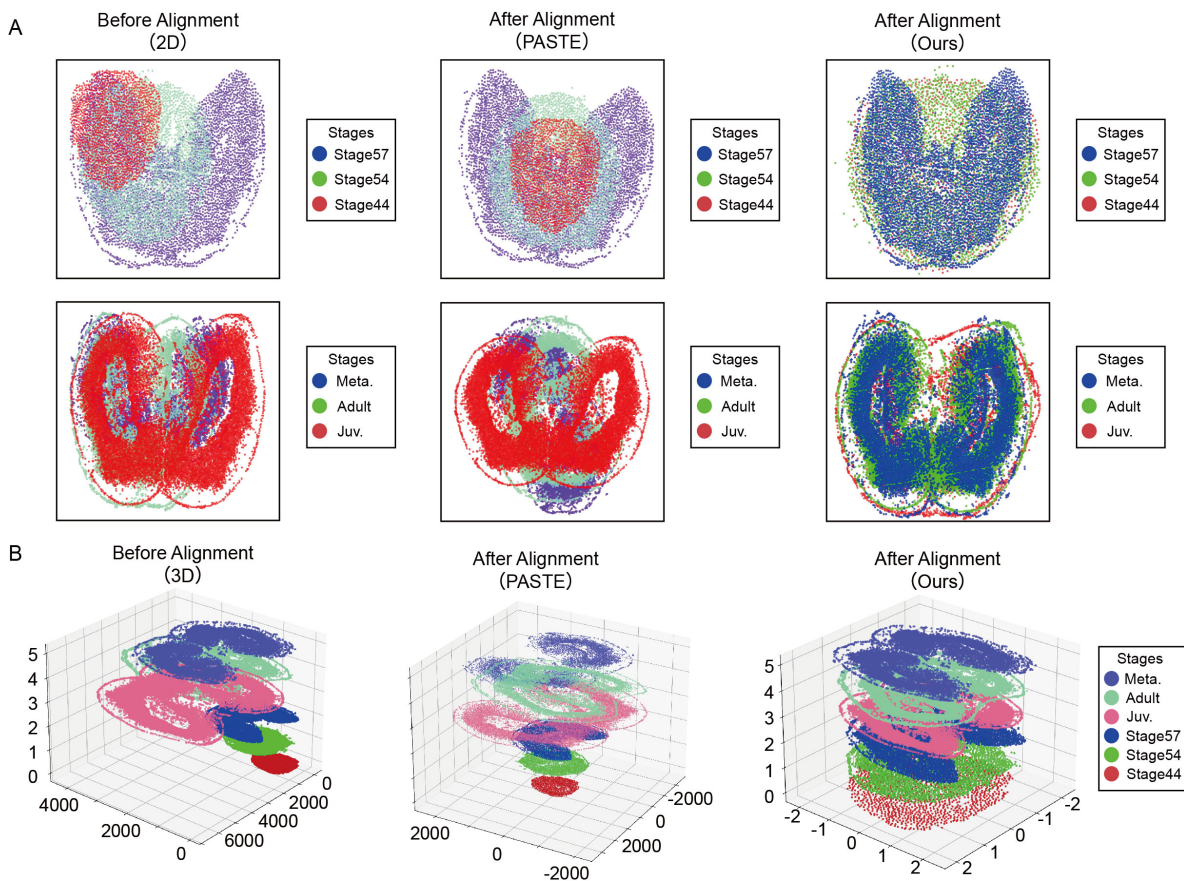


Fig. S7 Comparison of alignment performance between SpaTemporal and PASTE. **(A)** Comparison of 2D visualizations between SpaTemporal and PASTE in aligning axolotl embryos at early stages (St. 44–57, top) and late stages (Juv.–Meta., bottom). **(B)** Comparison of 3D visualizations between SpaTemporal and PASTE for aligning all developmental stages of axolotl embryos.

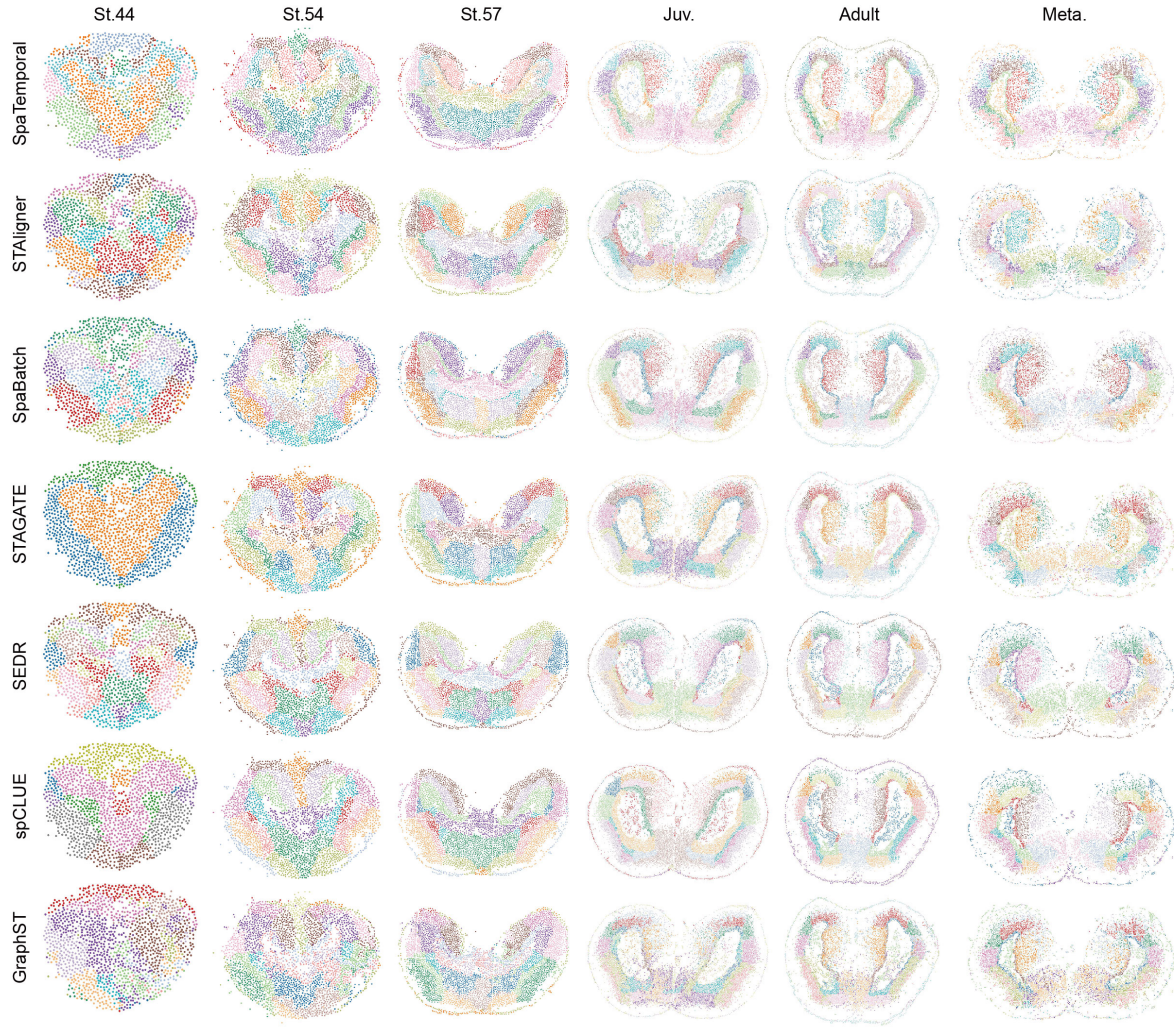


Fig. S8 Spatial domain identification results of SpaTemporal and baselines across all developmental stages of axolotl embryos.

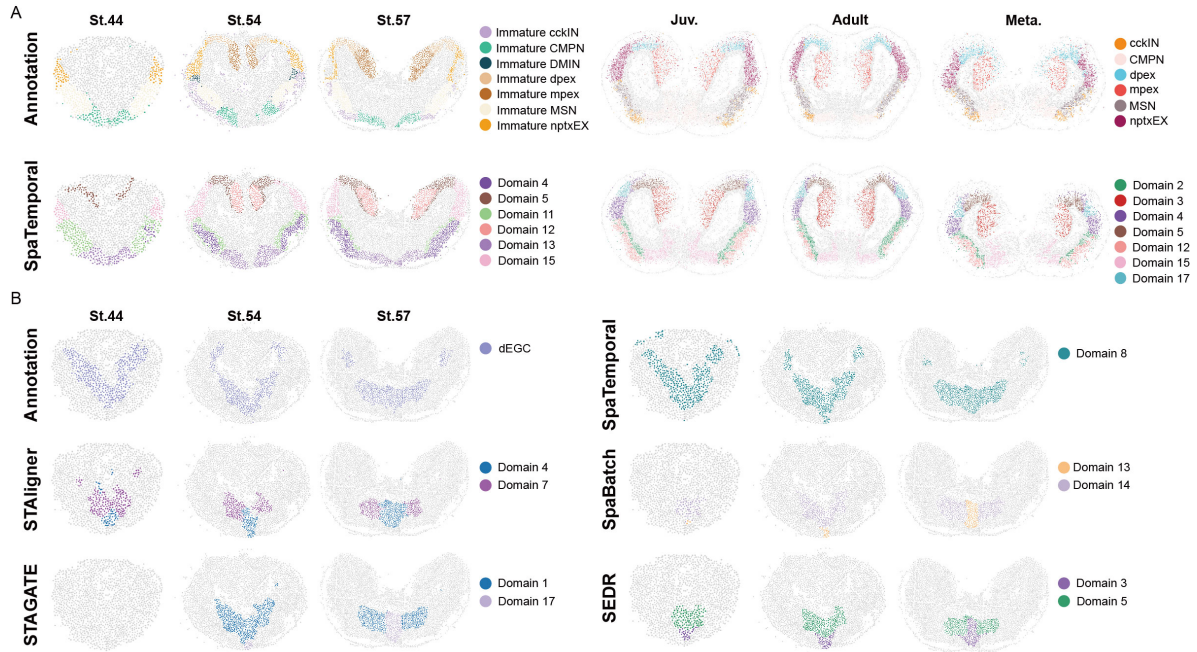


Fig. S9 Spatiotemporally mapping dEGC distribution and reconstructing the immature-to-mature tissue transformation. **(A)** SpaTemporal reconstructs the spatial shifts during the transition from early immature tissue to late mature tissue in axolotl embryos. **(B)** SpaTemporal precisely identifies the spatial location of dEGCs in early-stage axolotl embryos.

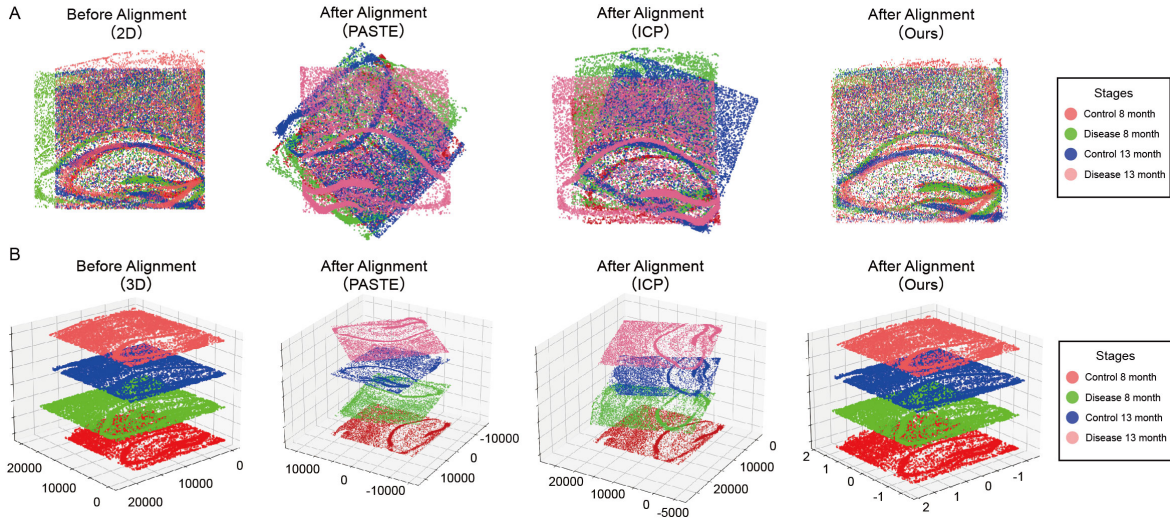


Fig. S10 Comparison of the (A) 2D and (B) 3D visualization performance among SpaTemporal, PASTE, and ICP in aligning Alzheimer's disease (AD) mouse data across different stages and pathological states.

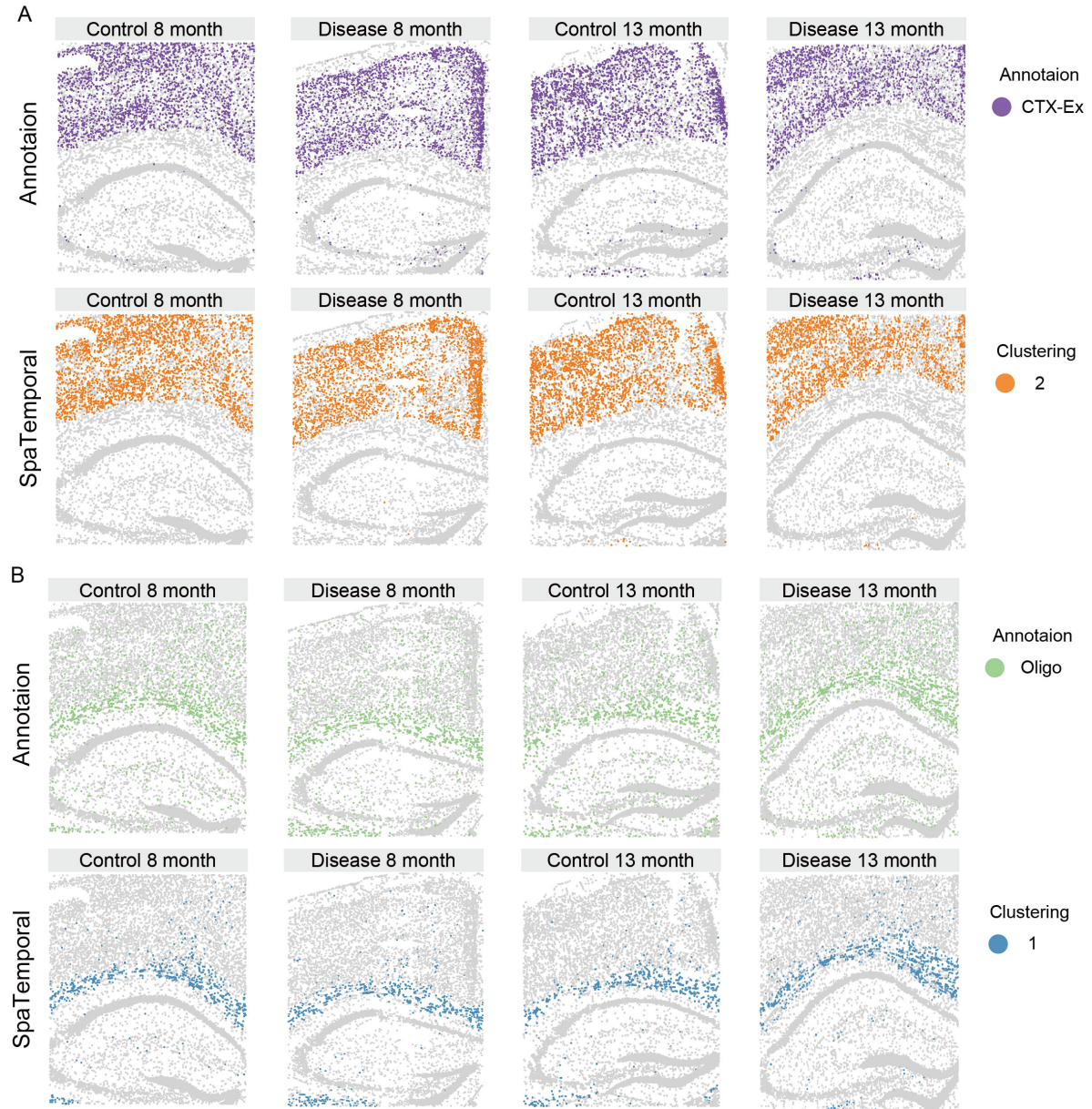


Fig. S11 SpaTemporal accurately identifies large-scale continuous spatial domains. **(A)** Spatial domain 2 identified by Spatiotemporal closely corresponds to the CTX-Ex annotated region. **(B)** Spatial domain 1 identified by Spatiotemporal closely corresponds to the Oligo annotated region.

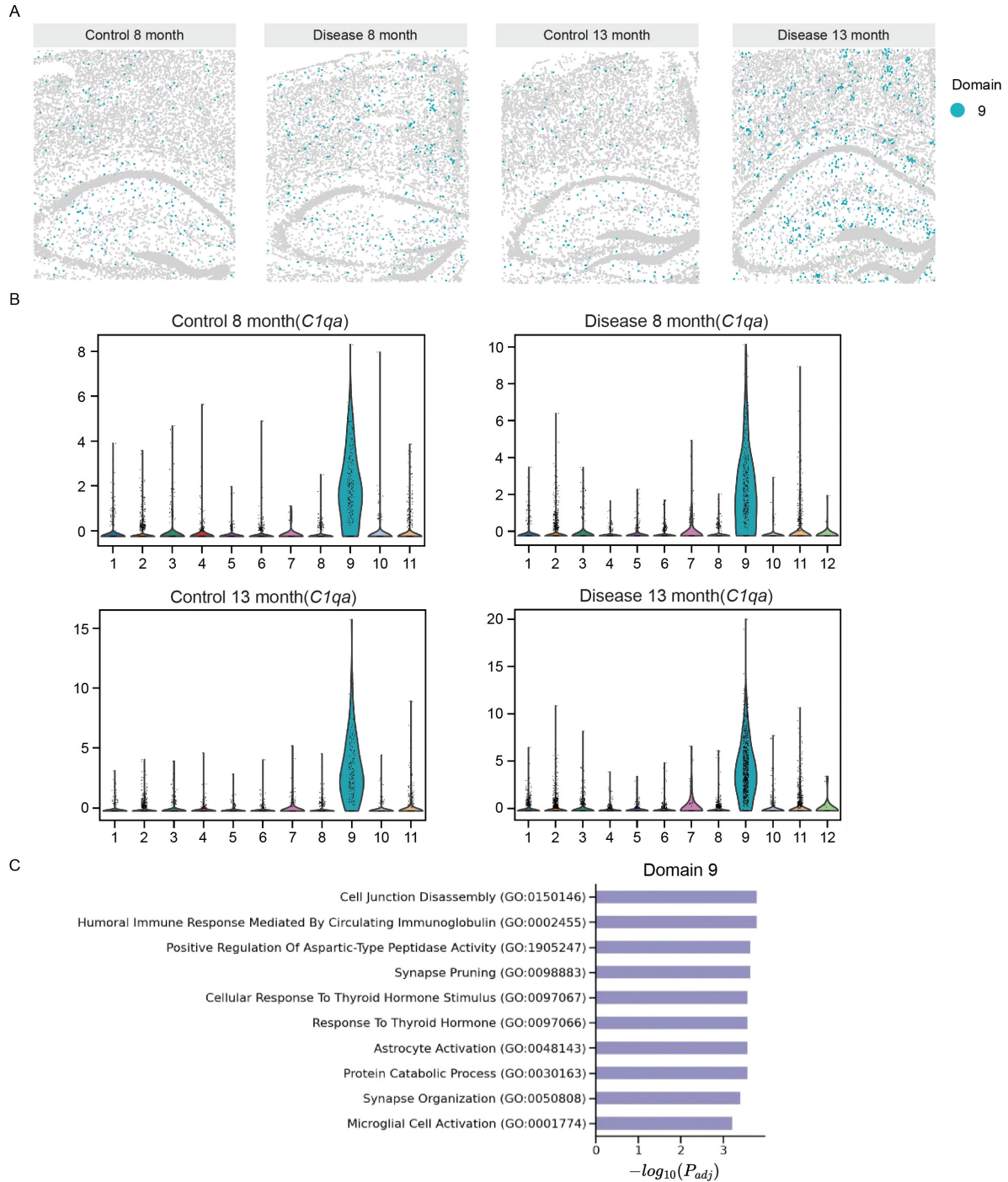


Fig. S12 Spatiotemporal changes of A β -Related spatial domains across different stages and states. **(A)** Spatial changes of A β -related spatial domains across different stages and states. **(B)** Changes in the expression levels of the marker gene *C1qa* across different stages and states. **(C)** GO enrichment analysis of genes associated with A β -related spatial domains.

2 Supplementary Tables

Table S1 Summary of the ST datasets used in this study.

Platform	Tissue	Section	Number of domains	Spots	Related figures
Stereo-seq	Mouse developing embryo dataset (MOE)	E9.5	12	5913	Figure 2 Figure S1-S3
		E10.5	18	8494	
		E11.5	19	30124	
Stereo-seq	Mouse brain datasets (MOB)	E12.5	18	9556	Figure 3 Figure S4-S6
		E13.5	18	12227	
		E14.5	18	17600	
		E15.5	18	17043	
		E16.5	18	16907	
Stereo-seq	Axolotl telencephalon	Stage44	15	1477	Figure 4 Figure S7-S9
		Stage54	15	2929	
		Stage57	15	4410	
		Juvenile	15	11698	
		Adult	15	8243	
		Metamorphosis	15	7441	
STARmap PLUS	Mouse Alzheimer's disease	Control 8 month	13	8506	Figure 5 Figure S10-S12
		Disease 8 month	13	8202	
		Control 13 month	13	8034	
		Disease 13 month	13	10372	

Table S2 Overview of comparative spatial domain identification methods.

Method	Methodology	Input Data	Downstream tasks	Link
Scanpy [Wolf et al., 2018]	Analysis Toolkit	Gene expression data	Spatial domain identification Visualization Trajectory inference Pseudo-temporal analysis	https://scanpy.readthedocs.io/en/stable/
STAGATE [Dong and Zhang, 2022]	Graph attention autoencoders	Spatial location data Gene expression data	Spatial domain identification Visualization Trajectory inference	https://github.com/zhanglabtools/STAGATE/
STAligner [Zhou et al., 2023]	Graph attention autoencoders	Spatial location data Gene expression data	Spatial domain identification Visualization Trajectory inference Temporal evolution Multi-slice integration	https://github.com/zhoux85/STAligner/
GraphST [Long et al., 2023]	Graph neural autoencoder	Spatial location data Gene expression data	Spatial domain identification Visualization Multi-slice integration	https://github.com/JinmiaoChenLab/GraphST/
SpaBatch [Niu et al., 2025]	Variational graph autoencoders	Spatial location data Gene expression data	Spatial domain identification Visualization Trajectory inference Multi-slice integration	https://github.com/wenwenmin/SpaBatch
SEDR [Xu et al., 2024]	Variational graph autoencoders	Spatial location data Gene expression data	Spatial domain identification Visualization Trajectory inference Multi-slice integration	https://github.com/HzFu/SEDR/
spCLUE [Wang et al., 2025]	Graph neural autoencoder	Spatial location data Gene expression data	Spatial domain identification Visualization Multi-slice integration	https://github.com/EnchantedJoy/spCLUE
PASTE [Zeira et al., 2022]	Principal Component Analysis	Spatial location data Gene expression data	Spatial domain identification Visualization Multi-slice alignment	https://github.com/raphael-group/paste

References

- Dong and Zhang, 2022. Dong, K. and Zhang, S. (2022). STAGATE: Deciphering spatial domains from spatially resolved transcriptomics with an adaptive graph attention auto-encoder. *Nature Communications*, 13(1):1739.
- Long et al., 2023. Long, Y., Ang, K. S., et al. (2023). GraphST: Spatially informed clustering, integration, and deconvolution of spatial transcriptomics with graphst. *Nature Communications*, 14(1):1155.
- Niu et al., 2025. Niu, J., Fang, D., et al. (2025). Spabatch: Deep learning-based cross-slice integration and 3d spatial domain identification in spatial transcriptomics. *Advanced Science*, 12(44):e09090.
- Wang et al., 2025. Wang, X., Li, W. V., et al. (2025). spclue: a contrastive learning approach to unified spatial transcriptomics analysis across single-slice and multi-slice data. *Genome Biology*, 26(1):177.
- Wolf et al., 2018. Wolf, F. A., Angerer, P., et al. (2018). SCANPY: large-scale single-cell gene expression data analysis. *Genome Biology*, 19:1–5.
- Xu et al., 2024. Xu, H., Fu, H., et al. (2024). SEDR: Unsupervised spatially embedded deep representation of spatial transcriptomics. *Genome Medicine*, 16(1):1–15.
- Zeira et al., 2022. Zeira, R., Land, M., et al. (2022). Alignment and integration of spatial transcriptomics data. *Nature Methods*, 19(5):567–575.
- Zhou et al., 2023. Zhou, X., Dong, K., et al. (2023). Integrating spatial transcriptomics data across different conditions, technologies and developmental stages. *Nature Computational Science*, 3(10):894–906.